

SOURCE CODE: -

```
#include <stdio.h>

#include <math.h>

// Recursive function representing
//  $T(n) = T(n-1) + n \log_2(n)$ 
double recurrence(int n)
{
    if (n <= 1)
        return 0;

    return recurrence(n - 1) + (n * log2(n));
}

int main()
{
    int n;

    double result;

    printf("Cybersecurity Threat Analysis System\n");
    printf("-----\n");
    printf("Enter the value of n: ");
    scanf("%d", &n);

    result = recurrence(n);

    printf("\nApproximate value of T(%d) = %.2lf\n", n, result);
    printf("\nTheoretical Time Complexity :  $\Theta(n^2 \log n)$ ");
    printf("\nSpace Complexity      :  $\Theta(n)$ ");
    return 0;
}
```

Output

```
Cybersecurity Threat Analysis System
-----
Enter the value of n: 8

Approximate value of T(8) = 85.53

Theoretical Time Complexity :  $\Theta(n^2 \log n)$ 
Space Complexity           :  $\Theta(n)$ 

=== Code Execution Successful ===
```